Lab 5: Forced vibration

Order: Reuse Lab 4 spring stiffnesses → measure damping with the motor OFF → measure motor travel → record the driven response.

A. Prepare the damped cart - motor OFF

1. Bring the **spring IDs and stiffnesses from Lab 4**. Record k_1 for the fixed-side spring and k_2 [N/m] for the motor-side spring. Do not repeat spring calibration unless you use different, uncalibrated springs.

2. Install the **1-2 magnetic dampers** to be used in the motor tests. Weigh the cart plus all installed moving parts, including hex rods and damper hardware. Record total moving mass **m [kg]** and magnet count.

3. Level the track. Attach both springs and keep both outer ends **stationary** for the damping test. Both springs must remain taut and the cart must stay clear of the end stops.

4. Open **Vernier Graphical Analysis → New Experiment → Sensor Data Collection**. Connect the cart by serial number. Display **Position (m) versus Time (s)**; set **25 samples/s**. With the cart resting at equilibrium, select **Position → Zero**.

B. Obtain the damping coefficient c - BEFORE the motor test

5. Click **Collect**. Pull the cart a small distance from equilibrium and release **without pushing**. Record several clean cycles of decreasing amplitude. Stop and use **document/file menu → Export → CSV → Save CSV**. Save as **Damped_Calibration.csv**. Note the clean start/end times after release.

6. Download **Lab 5 → MATLAB files → Complete bundle**; unzip and keep the three analysis .m files together. Run **Run_Lab5_Forced_Response.m**. Choose **1 = impulse/free decay**. Enter m [kg], k_1 and k_2 [N/m], then select Damped_Calibration.csv.

7. Enter the time/position column numbers (usually **[1 2]**; use **[2 3]** if an index column comes first). Enter the clean start/end times and choose **1 = meters**. Check that the fitted decay follows the measured points.

8. Open **Lab5_Results/Lab5_Impulse_Summary.csv**. Copy **c_est_Ns_m**: this is the effective damping coefficient **c [N s/m]** for this configuration. Save c and use **fn_theory_Hz** from the same table as the frequency-planning reference. Keep the same mass, springs and dampers during the motor sweep; changing them requires a matching decay measurement.

C. Measure the motor input

9. Keep the motor OFF while changing connections. Measure the drive endpoint at its two extreme positions using a ruler and record its **peak-to-peak travel**; target about **2-3 cm**. Reconnect the driven-side spring to the motor. The input peak amplitude is **Y = peak-to-peak travel / 2**. Example: 2.4 cm peak-to-peak is 0.024 m, so Y = 0.012 m.

Motor tests, tap tests and report

D. Record the motor-driven response

10. With the cart connected between the fixed spring and motor-driven spring, start the motor at low speed. Use the course supply range **2-11 V**. Measure actual speed with the tachometer. For one cycle per revolution, **f [Hz] = rpm/60**; voltage is not frequency.

11. Wait for approximately stable amplitude (often 1-2 minutes), then record **at least 5 settled cycles, preferably 10**. Ten cycles take 10/f seconds. Save one CSV per speed. Keep a startup-to-settled record at one speed. Note rpm, actual input peak-to-peak travel and settled start/end times.

12. Collect **3-5 frequencies below and 3-5 above** the resonance region. Keep mass, springs, magnets and input travel fixed. Do not dwell at resonance. Stop if motion grows excessively, springs go slack or the cart approaches an end stop.

13. Run **Run_Lab5_Forced_Response.m** again; choose **2 = harmonic sweep**. Enter the same m, both stiffnesses and your measured **c** from step 8. Select the motor CSV files. For each, enter its settled time window, meters, measured rpm and **input peak-to-peak travel in meters**. The code halves this travel; do not halve it before entering.

E. Record the three tap responses - motor OFF

14. Stop the motor. Remove magnetic dampers, record the resulting moving mass, and fix both outer spring ends so neither moves. Start from rest at equilibrium. Click Collect, give the cart body **one brief gentle horizontal finger tap**, then remove your hand. Record the decay. Repeat with light, medium and slightly stronger gentle taps; keep mass and springs unchanged.

15. Save **Tap_1.csv, Tap_2.csv and Tap_3.csv**. Analyze with **mode 1**, using the no-magnet moving mass and the same two stiffnesses. Select motion after contact ends. Compare amplitudes and measured frequencies. These tap levels are qualitative; force was not measured.

F. Keep these results for the combined Labs 4 + 5 report

1. Measured mass, spring IDs/stiffnesses, magnet count and damping c; include the calibration decay and fitted curve.
2. Three tap time histories together, and their spectra with peak frequencies labeled.
3. One motor startup-to-settled record and one settled data/fit/theory comparison.
4. Lab5_Frequency_Response.png: measured X/Y points overlaid on theory. Here X is fitted cart peak amplitude and Y is motor-endpoint peak amplitude. Include a compact table of f, X, Y and X/Y.

Use the outputs in **Lab5_Results**. Export figures directly. Explain departures from theory using your records; do not treat FFT peak height as X/Y. Keep raw CSVs and share the files with your group.

Before leaving: Check one damping fit, one tap analysis, one settled motor fit and the frequency-response plot.
Combined Labs 4 + 5 report: November 23, 11:59 PM.